Lecture 16

Plane wave propagation

From last time…

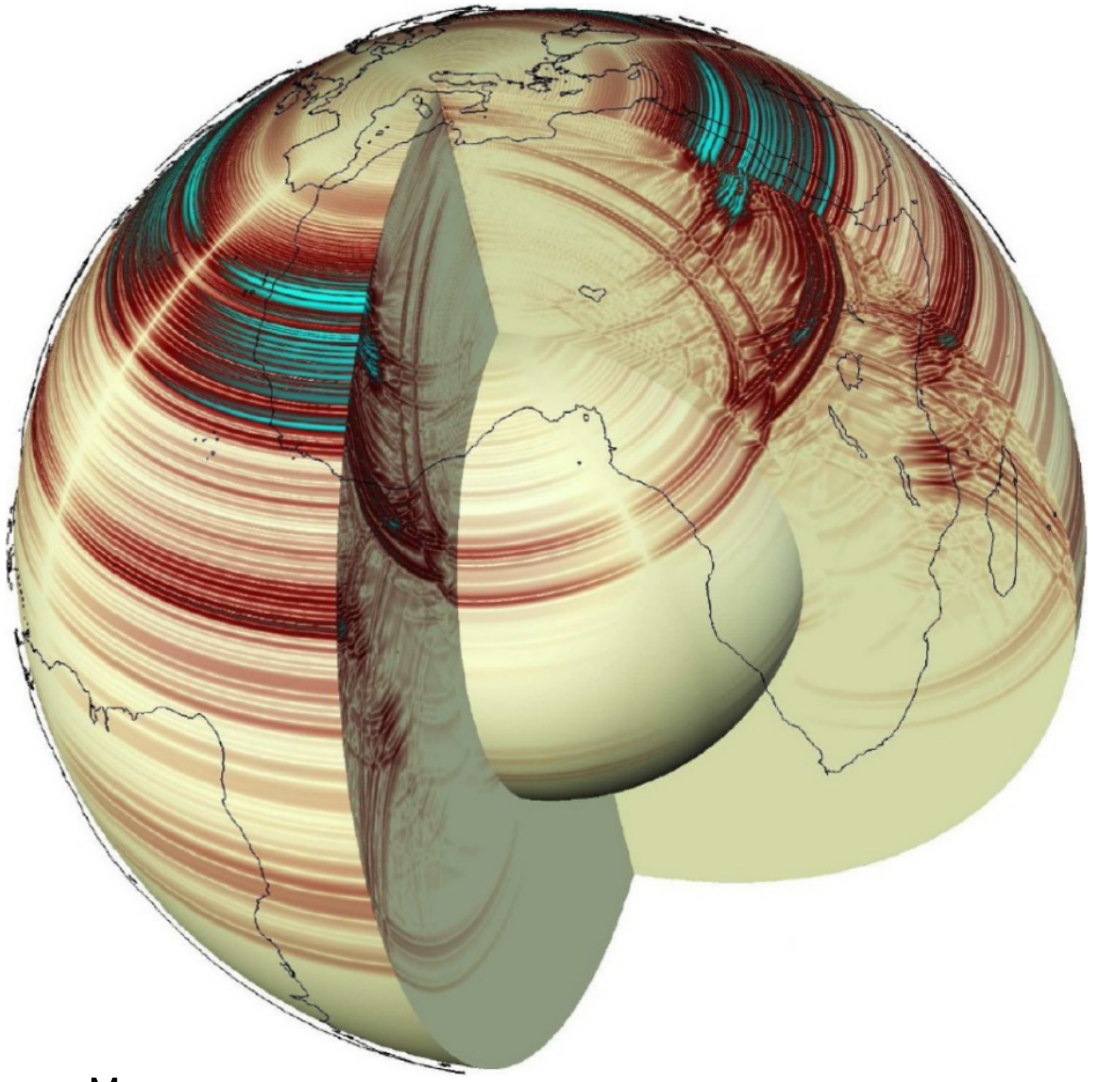
Linearised equations of motion including a stress glut:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = -\frac{\partial \overline{S}_{ij}}{\partial x_j},$$

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \overline{S}_{ij} \hat{n}_j.$$

Definition of the elastic tensor:

$$A_{ijkl}(\mathbf{x}) = \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}}(\mathbf{x}, \mathbf{I}).$$



Martin van Driel & Tarje Nissen-Meyer

The Christoffel equation

We consider an infinite and homogeneous body.

Look for a plane wave solution of the form:

$$u_i(\mathbf{x}, t) = f(t - p_j x_j) a_i.$$

The **slowness vector**, $\mathbf{p}$, sets the orientation and speed of the wavefronts.

The **polarisation vector**, $\mathbf{a}$, sets the direction of particle motion.

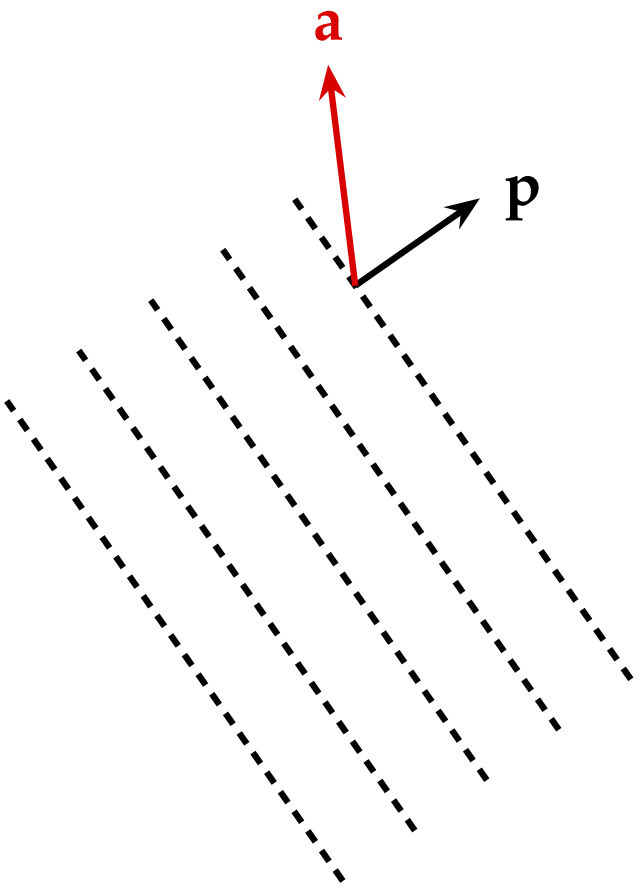
The **pulse shape** is set by f, and does not vary as the wave propagates – in this sense the waves are **non-dispersive** (but…).

Wavefronts:

$$t - p_j x_j = \text{constant}.$$

Phase speed:

$$c = \frac{1}{\|\mathbf{p}\|}.$$



For the plane wave ansatz:

$$\frac{\partial^2 u_i}{\partial t^2} = f''(t - p_j x_j) a_i,$$

$$\frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l}\right) = f''(t - p_m x_m) A_{ijkl} p_j p_l a_k.$$

The equations of motion are satisfied if and only if:

$$(\rho\delta_{ik} - A_{ijkl} p_j p_l) a_k = 0.$$

This is known as the **Christoffel equation** – it places a constraint on both the slowness and polarisation vectors.

It will be useful to define the **Christoffel matrix**, which is a function of $\mathbf{p}$:

$$\Gamma_{ik}(\mathbf{p}) = \frac{1}{\rho} A_{ijkl} p_j p_l.$$

The Christoffel equation can then be written:

$$[\mathbf{\Gamma}(\mathbf{p}) - \mathbf{I}]\mathbf{a} = \mathbf{0}.$$

This is a **homogeneous linear equation** for $\mathbf{a}$, that can have non-trivial solutions if and only if:

$$\det[\mathbf{\Gamma}(\mathbf{p}) - \mathbf{I}] = 0.$$

The roots of this equation define the **slowness surface**.

Reduction to an eigenvalue problem

From the definition of the Christoffel matrix, we have:

$$\mathbf{\Gamma}(\gamma\mathbf{p}) = \gamma^2\mathbf{\Gamma}(\mathbf{p}).$$

It is a **homogeneous function of order two**.

The hyperelastic symmetry implies that the Christoffel matrix is **symmetric**:

$$\Gamma_{ki}(\mathbf{p}) = \frac{1}{\rho}A_{kjil}p_j p_l = \frac{1}{\rho}A_{ilkj}p_j p_l = \frac{1}{\rho}A_{ijkl}p_j p_l = \Gamma_{ik}(\mathbf{p}).$$

Fixing the orientation of the slowness vector, we can write:

$$\mathbf{p} = \frac{1}{c}\hat{\mathbf{p}},$$

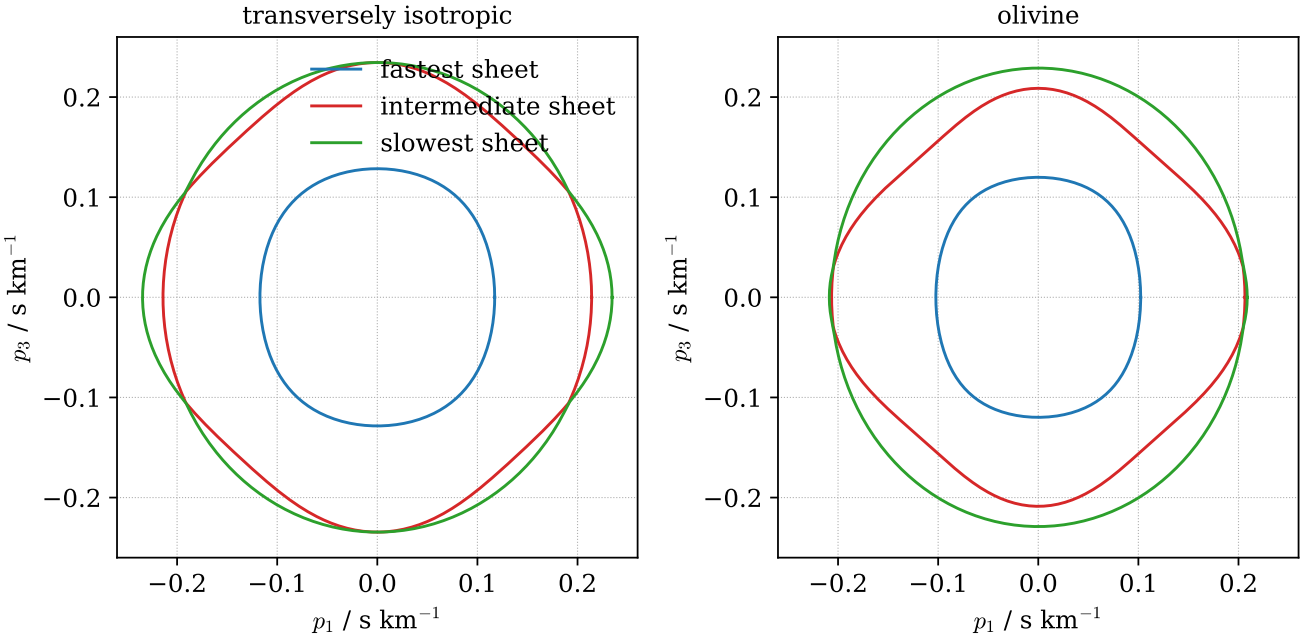
and hence the Christoffel equation reduces to:

$$\boldsymbol{\Gamma}(\hat{\mathbf{p}})\mathbf{a} = c^2\mathbf{a}.$$

This is a **symmetric eigenvalue problem**:

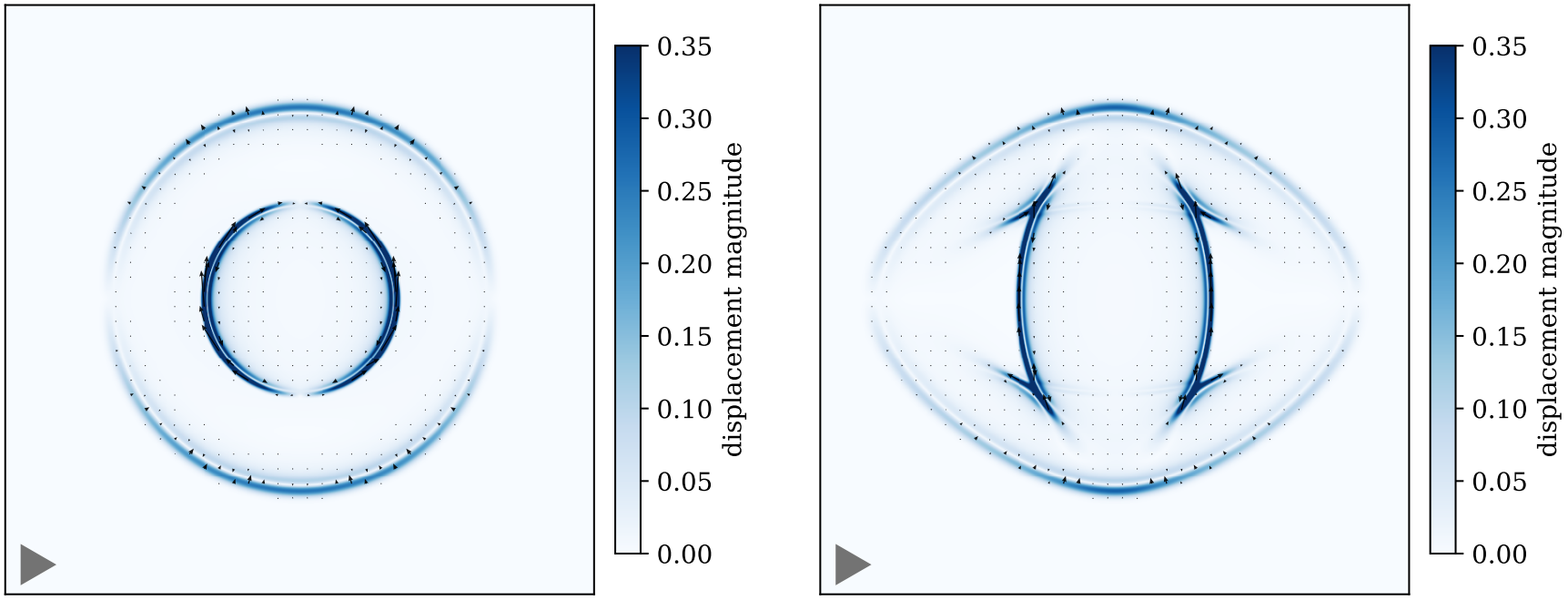
1) Squared phase speeds are **real**, generally **three distinct values**
2) Polarisation vectors for distinct squared phase speeds are **orthogonal**
3) Degenerate squared phase speeds are possible

The slowness surface has three sheets, which can touch along some directions:



$$[\|\mathbf{p}\|^2 c_1(\hat{\mathbf{p}})^2 - 1][\|\mathbf{p}\|^2 c_2(\hat{\mathbf{p}})^2 - 1][\|\mathbf{p}\|^2 c_3(\hat{\mathbf{p}})^2 - 1] = 0,$$

$$c_1(\hat{\mathbf{p}}) \leq c_2(\hat{\mathbf{p}}) \leq c_3(\hat{\mathbf{p}}).$$



A point force in an isotropic medium (left) and in a strongly anisotropic one (right). The quasi-P and quasi-S wavefronts are not circles, and the quasi-S front has cusps. These are two-dimensional solutions, with motion in the plane, so only two of the three sheets of the slowness surface appear.

Isotropic media

We have seen that the elastic tensor in an isotropic solid takes the form:

$$A_{ijkl} = \lambda \delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

Putting into the definition of the Christoffel matrix, we arrive at:

$$\begin{aligned}\Gamma_{ik}(\hat{\mathbf{p}}) &= \frac{\lambda+\mu}{\rho}\hat{p}_i\hat{p}_k + \frac{\mu}{\rho}\delta_{ik} \\ &= \frac{\lambda+2\mu}{\rho}\hat{p}_i\hat{p}_k + \frac{\mu}{\rho}(\delta_{ik} - \hat{p}_i\hat{p}_k).\end{aligned}$$

In the final expression, note the **orthogonal projections** parallel and perpendicular to the direction of the slowness vector.

Isotropic Christoffel matrix:

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{\lambda + 2\mu}{\rho}\hat{p}_i\hat{p}_k + \frac{\mu}{\rho}(\delta_{ik} - \hat{p}_i\hat{p}_k).$$

By inspection we identify an eigenvector and eigenvalue:

$$\mathbf{a} \propto \hat{\mathbf{p}}, \qquad \alpha = \sqrt{\frac{\lambda + 2\mu}{\rho}}.$$

These plane waves are known as **P-waves**:

- Particle motion along the propagation direction
- Phase speed the same in all directions
- Associated with a non-degenerate eigenspace

Isotropic Christoffel matrix:

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{\lambda + 2\mu}{\rho}\hat{p}_i\hat{p}_k + \frac{\mu}{\rho}(\delta_{ik} - \hat{p}_i\hat{p}_k).$$

By inspection we identify eigenvectors and an eigenvalue:

$$\mathbf{a} \perp \hat{\mathbf{p}}, \qquad \beta = \sqrt{\frac{\mu}{\rho}}.$$

These plane waves are known as **S-waves**:

- Particle motion perpendicular to the propagation direction
- Phase speed the same in all directions
- Associated with a two-fold degenerate eigenspace

Slightly anisotropic media

Anisotropy in the Earth







We consider a slightly anisotropic elastic tensor:

$$A_{ijkl} = \lambda \delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + s\,\delta A_{ijkl}.$$

Resulting Christoffel matrix:

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \Gamma^0_{ik}(\hat{\mathbf{p}}) + s\,\delta\Gamma_{ik}(\hat{\mathbf{p}}), \qquad \delta\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{1}{\rho}\delta A_{ijkl}\hat{p}_j\hat{p}_l.$$

Eigenvalue problem to be solved:

$$[\mathbf{\Gamma}^0(\hat{\mathbf{p}}) + s\,\delta\mathbf{\Gamma}(\hat{\mathbf{p}})]\mathbf{a} = c^2\mathbf{a}.$$

Apply **non-degenerate** perturbation theory:

$$c = \alpha + s\,\delta\alpha + \cdots,$$
$$\mathbf{a} = \hat{\mathbf{p}} + s\,\delta\mathbf{a} + \cdots.$$

First-order term in eigenvalue problem:

$$\boldsymbol{\Gamma}^0(\hat{\mathbf{p}})\delta\mathbf{a} + \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{p}} = \alpha^2\delta\mathbf{a} + 2\alpha\,\delta\alpha\,\hat{\mathbf{p}}.$$

Through a standard argument, we arrive at:

$$\delta\alpha = \frac{1}{2\alpha}\hat{\mathbf{p}}\cdot\delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{p}}.$$

Polarisation **almost aligned** with propagation direction, phase speed now **varies with the orientation**. This is a **quasi P-wave**.

Apply **degenerate** perturbation theory:

$$c = \beta + s\,\delta\beta + \cdots,$$
$$\mathbf{a} = a_1\hat{\mathbf{t}}_1 + a_2\hat{\mathbf{t}}_2 + s\,\delta\mathbf{a} + \cdots.$$

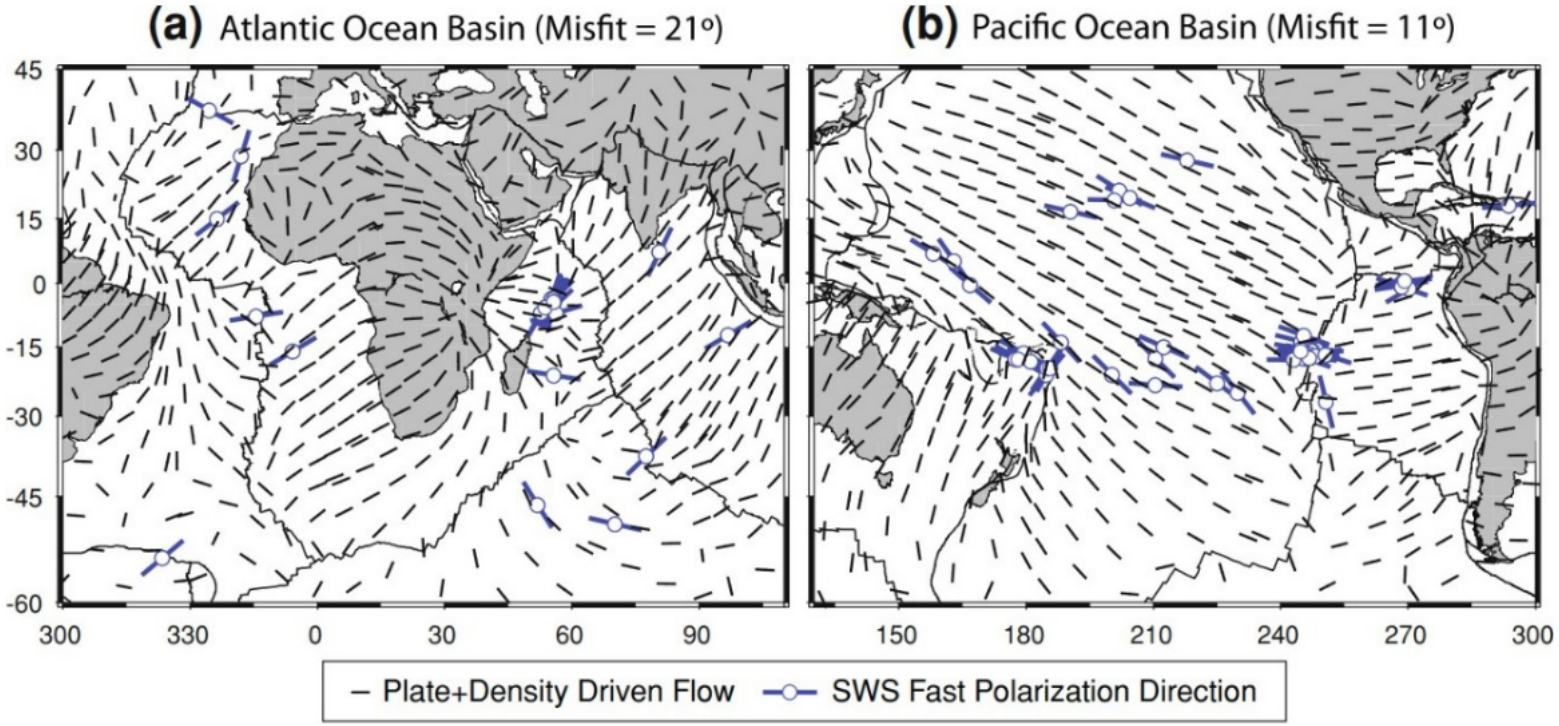
Leads to a 2×2 symmetric eigenvalue problem:

$$\begin{pmatrix} \hat{\mathbf{t}}_1 \cdot \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{t}}_1 & \hat{\mathbf{t}}_1 \cdot \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{t}}_2 \\ \hat{\mathbf{t}}_2 \cdot \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{t}}_1 & \hat{\mathbf{t}}_2 \cdot \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{t}}_2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 2\beta\,\delta\beta \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}.$$

Solution gives **two perturbed phase speeds** which are usually distinct – the initial **degeneracy is split**.

Zeroth-order eigenvectors orthogonal to each other and to the propagation direction; the first-order correction will have a component along the propagation direction.

These are **quasi S-waves**, and this behaviour is known as **shear wave splitting**.



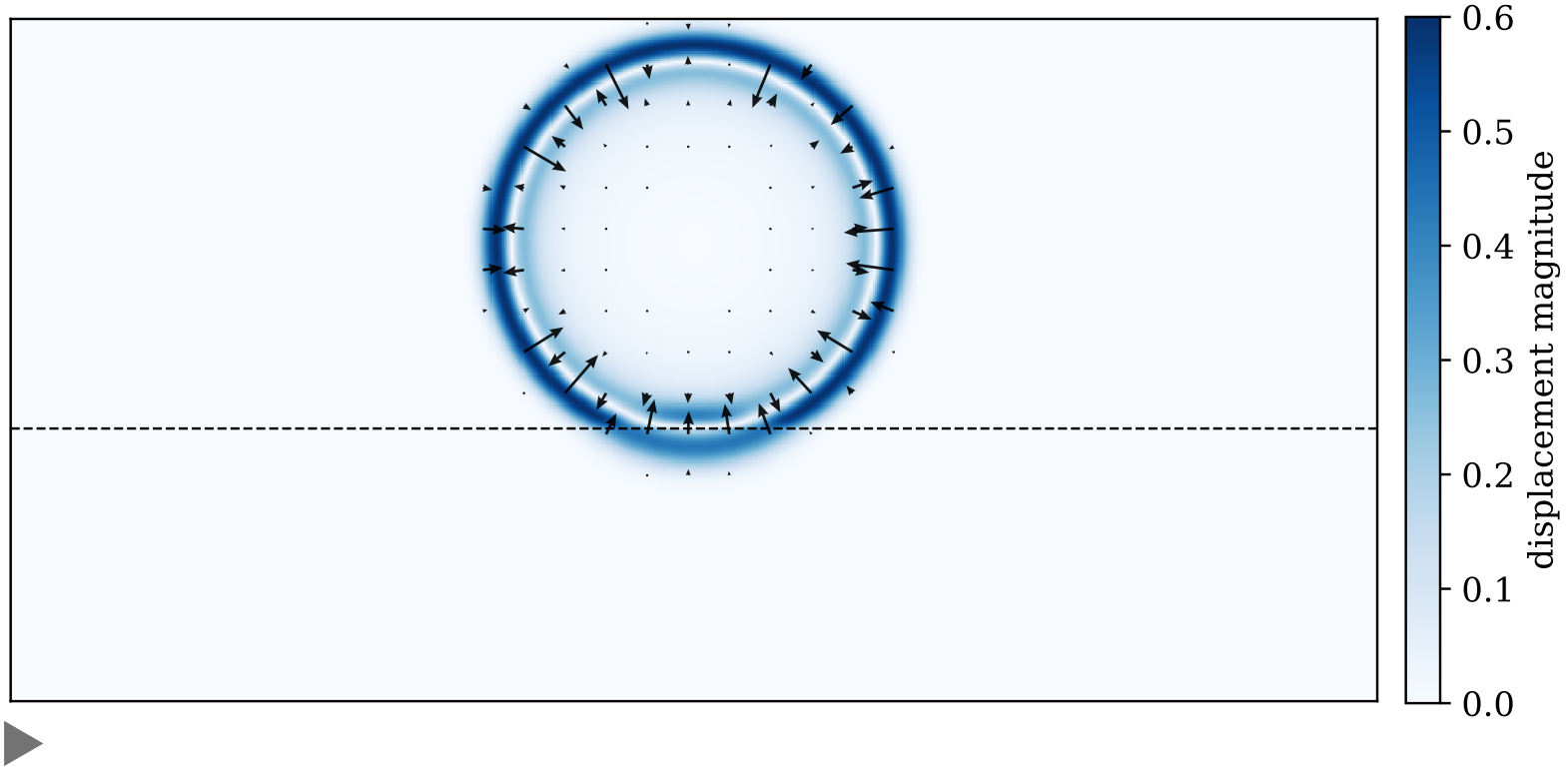
Convection in the mantle creates large-scale **alignment** of anisotropic mineral grains with the **flow direction** – produces an effective seismic anisotropy. Example here shows the relation between shallow anisotropy and plate motion.

Long & Silver (2009)

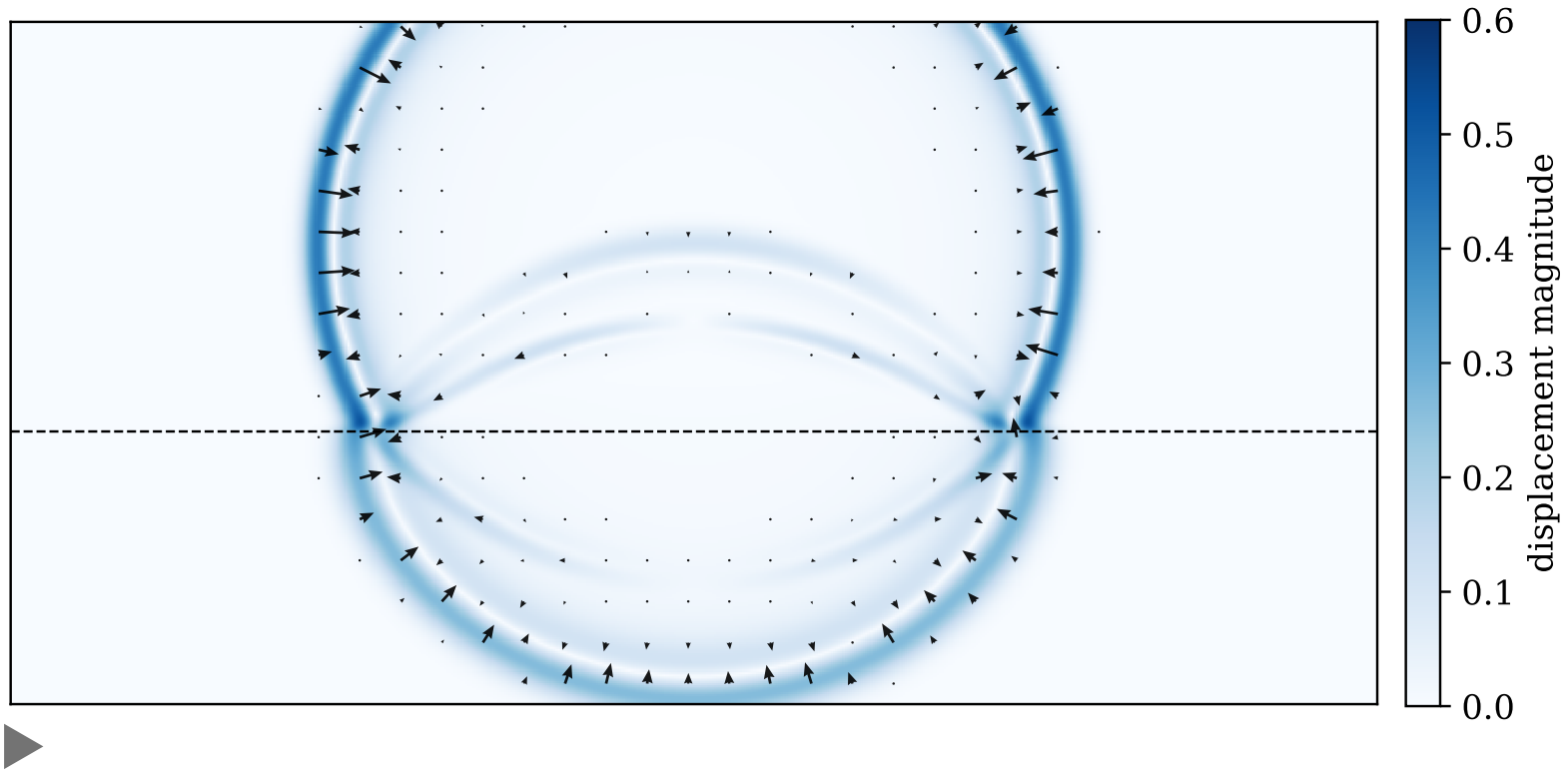
The **interaction of plane waves with boundaries** is not treated quantitatively in this course.

The usual **reflections** and **transmissions** of waves occur when a plane elastic wave interacts with a boundary.

The only interesting feature is the **conversion** between different wave types. For example, in an isotropic medium, an incident P-wave can produce reflected and transmitted P- and S-waves.



A P-wave from a point source reaching a boundary with a faster medium.



Reflected and transmitted P- and S-waves, and head waves along the boundary. Arrows show the particle motion.

What you need to know and be able to do

(i) How to derive the Christoffel equation from the linearised equations of motion.

(ii) Reduction of the problem to a symmetric eigenvalue problem, and a qualitative description of its solution.

(iii) How to solve the Christoffel equation for an isotropic solid, and to deduce the properties of plane P- and S-waves.

(iv) How to use first-order perturbation theory to study the effect of slight anisotropy on plane waves.